AIC-607 Brain-Computer Interface

EEG Signal Processing Report

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Part 1: EEG Data Collection

1.1 Subject and Environment

The EEG data was recorded from a single, healthy adult who is one of the group members. The recording took place in the quiet room which made the environment sufficiently quiet for the recording. This setting was convenient and still provided the low noise conditions necessary for good quality EEG data. The subject sat comfortably in a chair with his feet flat on the floor and his hands resting on his lap. He maintained a relaxed but alert posture throughout the recording to help reduce muscle tension and movement, both of which can create unwanted artifacts in the EEG signal.

1.2 Equipment and Setup

For this experiment, the Unicorn EEG Headset was used, which has eight channels placed at positions Fz, C3, Cz, C4, Pz, PO7, Oz, and PO8 according to the standard 10-20 system. The headset samples data at 250 times per second, which is sufficient to capture the brain waves, such as alpha and beta rhythms. The reference and ground electrodes are built into the device and positioned on the earlobes.

The headset was first positioned on the subject's head and adjusted to ensure that each electrode was correctly aligned according to the 10–20 system. Following placement, conductive gel was applied to each electrode. This sequence was maintained to ensure accurate electrode positioning prior to gel application. A small amount of gel was injected into each electrode until contact between the gel and the scalp was observed through the central opening of the electrode. The application of conductive gel served to reduce the impedance between the electrode and the scalp; without it, electrical resistance would remain high, resulting in weak and noisy signals.

After gel application to all eight electrodes, impedance levels were assessed using the Unicorn Recorder software. The software provided a color-coded indication for each electrode, and all channels were verified to display green, signifying that impedance levels were within an acceptable range. Recording was subsequently initiated. The software further enabled data acquisition and storage in CSV format, which facilitated subsequent processing in Python using the MNE library.

1.3 Recording Protocol

The recording session was conducted over a total duration of five minutes and was divided into five blocks of sixty seconds each. A timer was used to monitor the duration of each block, and a predefined set of instructions was followed for each experimental condition. Prior to the commencement of recording, the instructions were read aloud to the participant to ensure a clear understanding of the tasks associated with each block. This preparatory step was undertaken to promote consistency in task performance throughout the session.

During the first block, corresponding to the eyes-open resting condition, the participant was instructed to fixate on a stationary point displayed on a laptop screen, minimize unnecessary blinking, and remain as still as possible. For the second block, representing the eyes-closed resting condition, the participant was instructed to close their eyes and refrain from engaging in directed thought.In the third block, which involved a mental arithmetic task, the participant was instructed to keep their eyes closed and silently perform serial subtraction of seven from three hundred (e.g., 300, 293, 286, and so forth). It was emphasized that no verbalization or lip movement should occur, as such muscle activity could introduce artifacts into the recorded signals.The fourth block consisted of a repetition of the eyes-open resting condition, while the fifth block involved a repetition of the eyes-closed resting condition.

At the conclusion of each sixty-second interval, a gentle tap on the participant's shoulder was used as a non-verbal cue to indicate the transition to the subsequent block. Verbal communication was deliberately avoided during the recording period to prevent the introduction of auditory evoked potentials and other sources of noise into the data.

1.4 Resulting Data

The Unicorn Recorder software saved the raw EEG data as a CSV file named UnicornRawDataRecorder_24_03_2026_11_36_120.csv. The file contains eight columns of EEG data, one for each channel, along with a timestamp column. The total recording duration was three hundred and one seconds, slightly over three hundred seconds, which gave enough data for the five blocks.

Part 2: Signal Cleaning

Step 1: Loading the recording

The CSV file is loaded and the 8 EEG channels are pulled out. The values are converted from microvolts to volts because that is what MNE uses. The channels are named, given a sampling rate of 250 Hz, and placed on a standard scalp map. The first 3.97 seconds are cut off, then a PSD plot is made.The first few seconds are cut because the headset uses them to calibrate a sensor, not to record brain activity. The scalp map is needed so later steps know where each electrode sits. The PSD is a first look at signal quality before any cleaning.

The recording goes from 301 seconds down to about 297 seconds. The PSD will show lots of slow drift at low frequencies and a tall spike at 50 Hz from electrical noise. The output confirms 8 channels, 250 Hz sampling, and 297.03 seconds of data. The "bads: []" line just means no channels have been flagged yet. Everything loaded correctly.

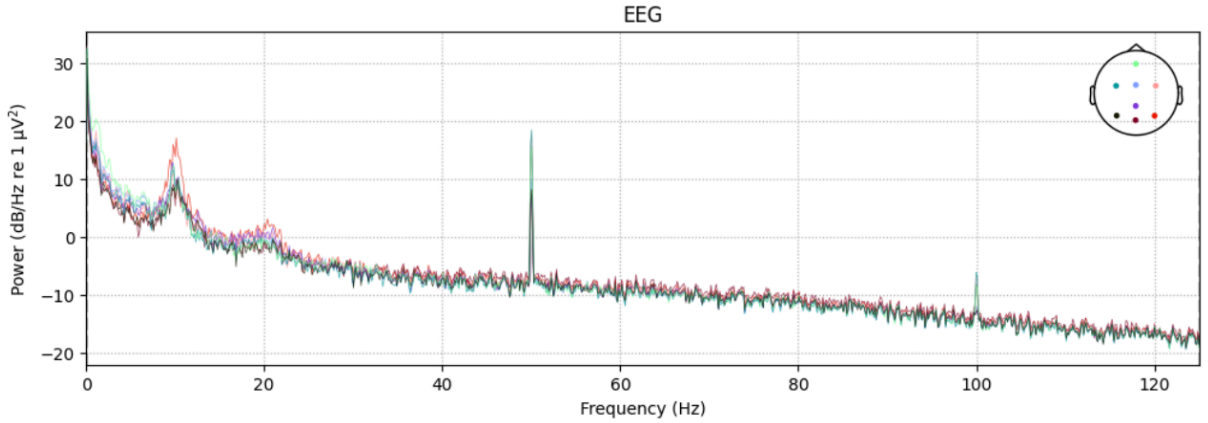


Figure 1. Power Spectral Density (PSD) of the raw, unfiltered EEG data.

Step 2: Checking the scalp map

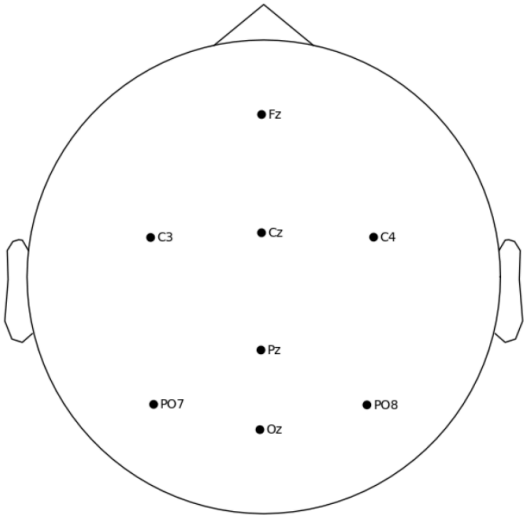


Figure 2. Scalp Map

A schematic representation was generated to illustrate the spatial placement of each electrode on the scalp, serving as a rapid visual verification of the setup. Any incorrect configuration would be reflected by electrodes appearing in anatomically inconsistent positions.

It was observed that no signal characteristics were altered during this process, as the procedure solely involved visualization. The resulting plot was expected to display Fz in the frontal region, C3 and C4 laterally, Cz at the vertex, Pz in the parietal region, and PO7, Oz, and PO8 in the occipital region. A symmetrical arrangement of these electrode positions was taken as confirmation that the setup had been correctly implemented.

Step 3: Applying filters

Three filters were applied to the EEG data. A high-pass filter at 1 Hz was used to remove slow drift, a low-pass filter at 40 Hz was applied to eliminate high-frequency noise, and a notch filter at 50 Hz was implemented to suppress interference from the electrical power supply. Subsequently, the power spectral densities (PSDs) of both the raw and filtered signals were plotted for comparison.

Raw EEG data are typically affected by several sources of noise, including slow drift, high-frequency muscle artifacts, and a 50 Hz electrical hum. The application of the aforementioned filters was intended to attenuate these unwanted components.Following filtering, a reduction in both very low- and high-frequency components was observed, while the prominent 50 Hz peak was substantially diminished. The intermediate frequency range, which contains the primary neural signals, was preserved with improved clarity.

In the resulting plot, the dashed lines representing the raw data exhibited elevated power at both low- and high-frequency extremes, whereas the solid lines corresponding to the filtered data showed significantly reduced power in these regions. The pronounced peak at 50 Hz in the raw data was effectively removed in the filtered signal. In contrast, the mid-frequency range demonstrated close overlap between raw and filtered PSDs, indicating that the filtering process successfully removed noise while preserving the underlying brain activity.

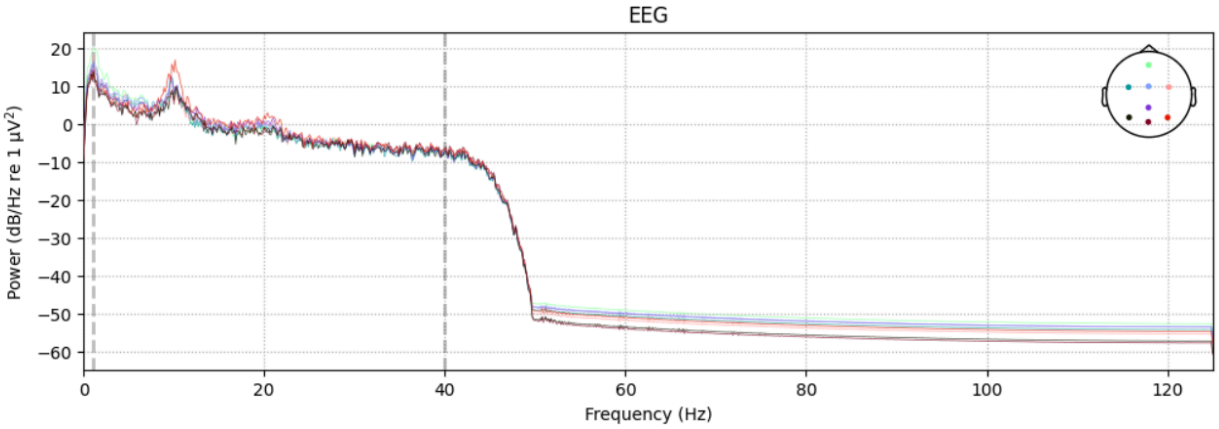
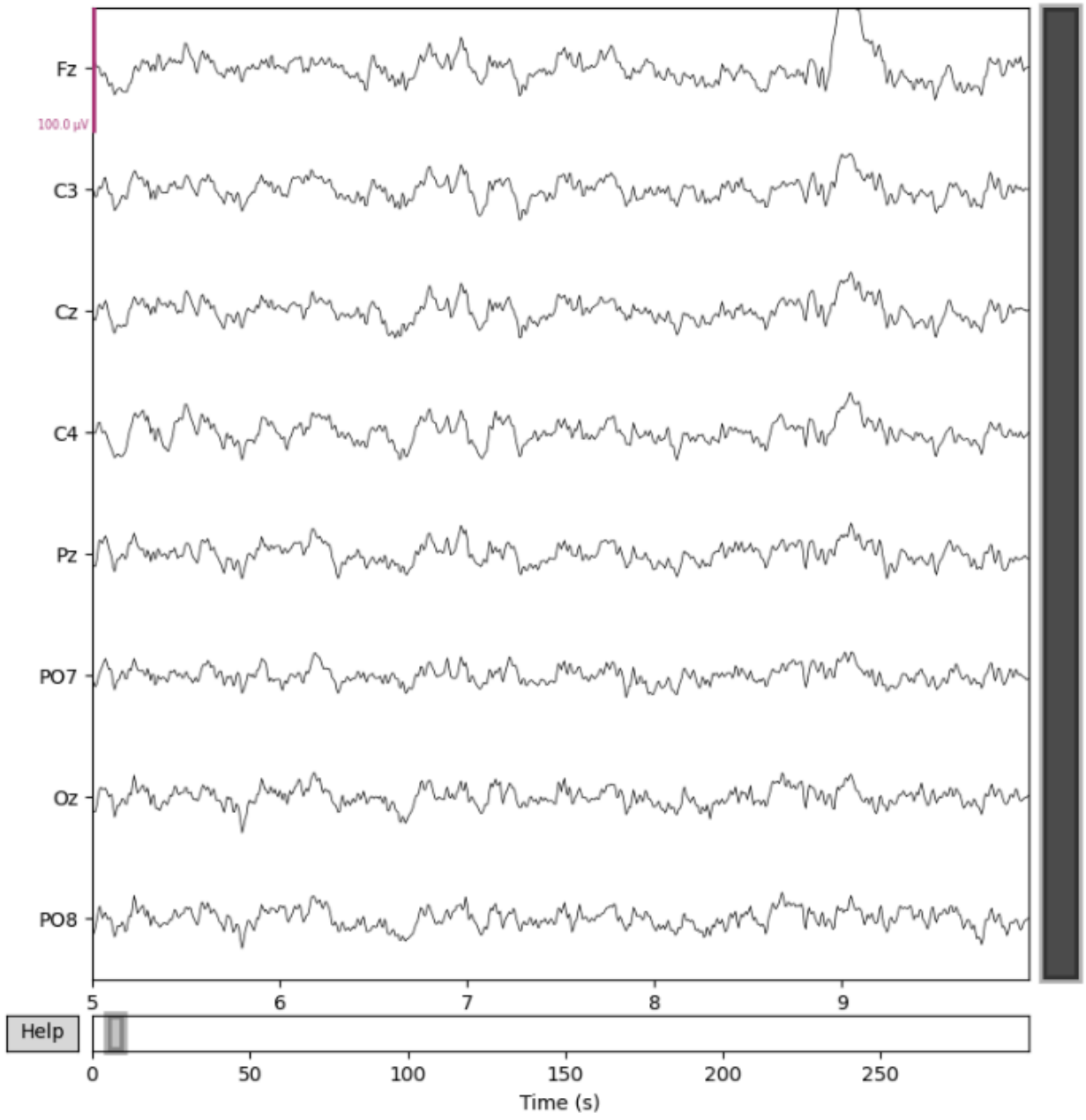


Figure 3. PSD after filtering.

Step 4: Checking for bad channels

A five-second segment of the filtered EEG signal was visualized across all eight channels. This step was performed to enable inspection of the signal quality following preprocessing. Despite prior filtering, certain channels may still exhibit issues, such as flat-line behavior or excessive noise. Visual inspection by a human observer is considered the most reliable method for identifying such abnormalities.No modifications were made to the signal during this stage, as the procedure served solely as a quality control check. The output indicated “Bad channels: []”, signifying that no channels were identified as faulty. Consequently, all eight channels were retained for subsequent analysis.




```
Bad channels: []
```

Figure 4. Visual inspection of the filtered EEG signal across all 8 channels.

Step 5: Cutting into epochs and labeling

The continuous EEG recording was segmented into five task-specific blocks, each with a duration of sixty seconds. Subsequently, each block was further divided into shorter segments of two seconds, referred to as epochs. Each epoch was labeled according to the corresponding task

condition. Continuous EEG data are not well-suited for direct analysis; therefore, segmentation into fixed-length, labeled epochs is required for compatibility with standard analytical methods. A duration of two seconds was selected as it provides sufficient temporal resolution for reliable frequency estimation while yielding an adequate number of samples per condition. Following this process, the dataset was structured as a three-dimensional array with dimensions corresponding to epochs, channels, and timepoints. Each epoch contained 500 samples, consistent with a sampling rate of 250 Hz over a two-second interval.

```
eyes_open_1         : 59 epochs
Not setting metadata
59 matching events found
No baseline correction applied
0 projection items activated
Using data from preloaded Raw for 59 events and 500 original time points ...
0 bad epochs dropped
eyes_closed_1       : 59 epochs
Not setting metadata
59 matching events found
No baseline correction applied
0 projection items activated
Using data from preloaded Raw for 59 events and 500 original time points ...
0 bad epochs dropped
mental_arithmetic   : 59 epochs
Not setting metadata
59 matching events found
No baseline correction applied
0 projection items activated
Using data from preloaded Raw for 59 events and 500 original time points ...
0 bad epochs dropped
eyes_open_2         : 59 epochs
Not setting metadata
56 matching events found
No baseline correction applied
0 projection items activated
Using data from preloaded Raw for 56 events and 500 original time points ...
0 bad epochs dropped
eyes_closed_2       : 56 epochs
Not setting metadata
292 matching events found
No baseline correction applied
Total epoch: 292
Data shape: (292, 8, 500)
```

Figure 5: Epoching and Labeling Summary.

The resulting output indicated 59, 59, 59, 59, and 56 epochs for the respective blocks, yielding a total of 292 epochs with a final shape of (292, 8, 500). The reduced number of epochs in the final block was attributed to the recording duration being insufficient to accommodate the final two-second segment. No epochs were excluded at this stage.

Step 6: Rejecting bad epochs

Epochs in which the signal amplitude exceeded ±100 microvolts were rejected, and the corresponding labels were updated accordingly.

Although earlier preprocessing steps reduce noise, transient artifacts arising from eye blinks, jaw movements, or other sources may still be present. The application of an amplitude threshold serves as a straightforward and efficient method for eliminating such contaminated segments. As a result of this procedure, the total number of epochs was reduced; however, the remaining data were of substantially higher quality and more suitable for subsequent analysis.

```
42 bad epochs dropped
Epochs after rejection : 250
Counts per block       :
  eyes_open_1          : 46
  eyes_closed_1        : 55
  mental_arithmetic    : 55
  eyes_open_2          : 48
  eyes_closed_2        : 46
```

Figure 6: Epoch Rejection Based on Peak-to-Peak Amplitude.

The results indicated that 42 epochs were rejected, leaving a total of 250 epochs for further analysis. The majority of rejections were associated with the Fz channel, which is consistent with its frontal placement and sensitivity to eye-blink artifacts. The PO8 channel also contributed to a smaller number of rejections, likely due to muscle activity in the posterior region of the head.

The distribution of the remaining epochs across the five blocks was 46, 55, 55, 48, and 46, respectively, indicating a relatively balanced dataset. Notably, the eyes-open conditions exhibited the highest number of rejected epochs, which aligns with the increased likelihood of blinking during these periods.

Step 7: Fitting ICA

Independent Component Analysis (ICA) was applied to the filtered EEG data, with the number of components selected to account for 99% of the variance in the signal. Rejecting entire epochs can result in unnecessary data loss, particularly when the noise arises from recurring artifacts such as eye blinks. ICA addresses this limitation by decomposing the EEG signal into statistically independent components, which represent underlying sources. Some of these components correspond to neural activity, while others capture artifacts such as ocular or muscular interference. This separation enables targeted removal of artifact-related components without discarding entire segments of data. At this stage, no modifications were made to the original signal, as ICA was used solely to perform the decomposition into independent components.

```
Fitting ICA to data using 8 channels (please be patient, this may take a while)
Selecting by explained variance: 7 components
Fitting ICA took 0.4s.
ICA fitted.
```

Figure 7: Fitting ICA

The results indicated that 7 components were retained out of a possible 8, and the ICA fitting process required approximately 0.4 seconds. Given the relatively small number of channels, the capacity of ICA to fully separate all underlying sources is inherently limited. Nevertheless, retaining 7 components is sufficient to capture dominant artifacts, such as those associated with eye blinks.

Step 8: Looking at the ICA components

Three types of visualizations were generated: scalp topographies for each independent component, temporal activation patterns of the components, and a detailed inspection of component 0. Prior to the removal of any components, it was necessary to examine these visualizations to identify potential artifacts. Components associated with eye blinks are typically characterized by strong activation in the frontal regions of the scalp and exhibit large, sharp deflections in their time courses, often occurring at a frequency of approximately one to two events per second. No modifications were made to the EEG signal at this stage, as these visualizations were intended solely to inform the subsequent identification and removal of artifact-related components.

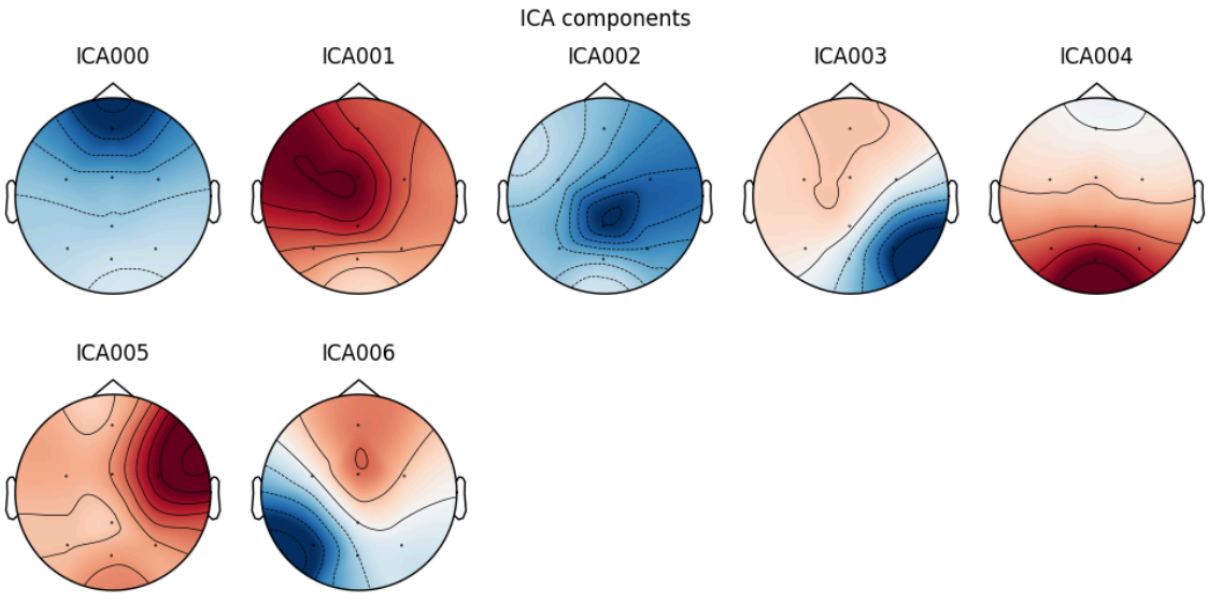


Figure 8. ICA component topographies.

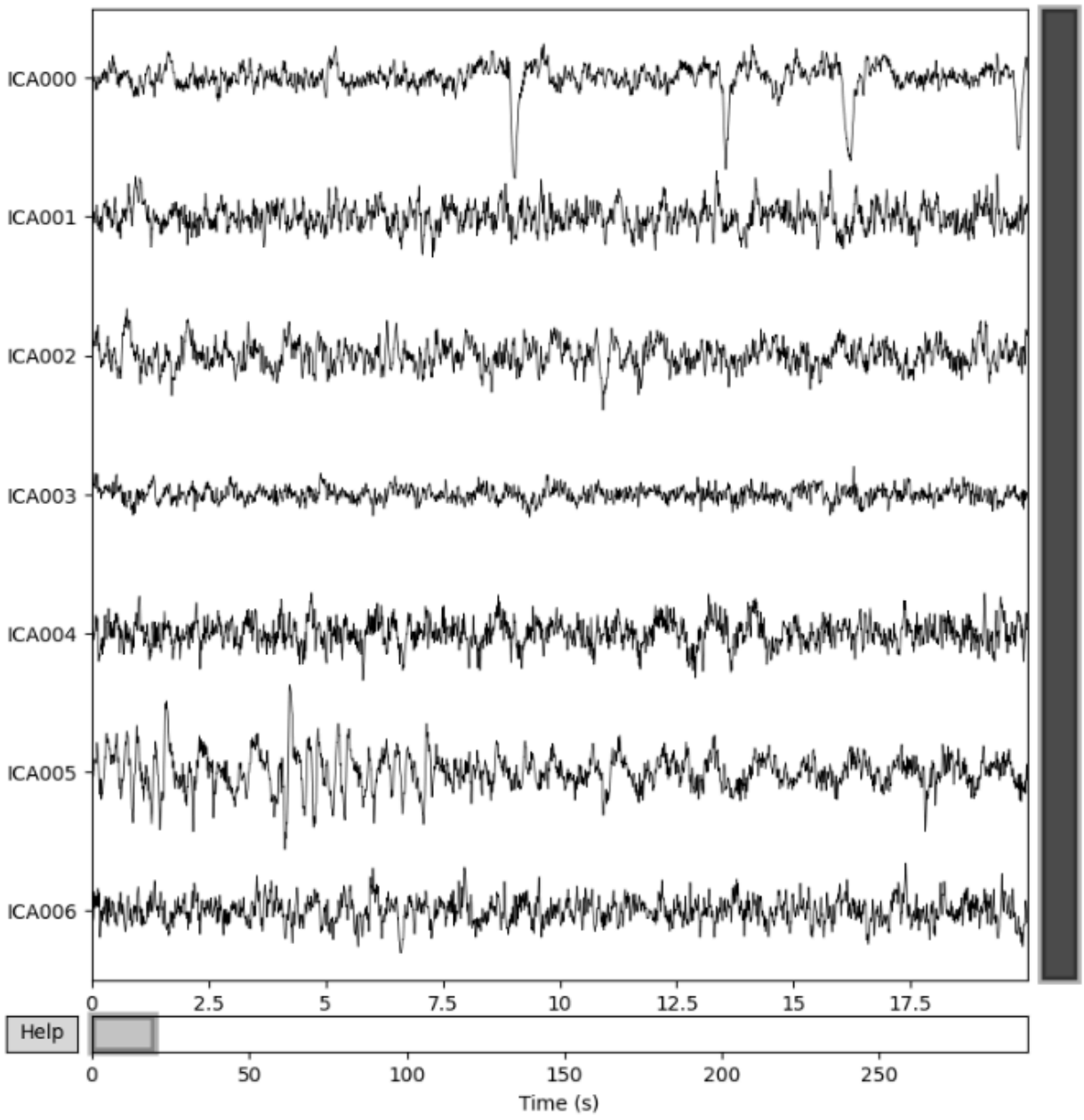


Figure 9. Time courses of ICA components.

Based on the visual inspection of the component plots, components 0 and 1 were identified as artifacts. The corresponding scalp topographies exhibited strong activation in the frontal regions, while their temporal profiles displayed prominent, spike-like patterns characteristic of eye-blink activity.

Step 9: Removing the bad components

Components 0 and 1 were designated for removal, and ICA was subsequently applied to a copy of the dataset to reconstruct the EEG signal with these components excluded. To evaluate the effect of this procedure, the Fz channel was examined before and after component removal, and the power spectral density (PSD) at the Oz channel was compared across the same conditions.

Identification of artifact-related components alone is insufficient; the removal must be explicitly applied to the data. The Fz channel was selected for inspection due to its proximity to the frontal region, where eye-blink artifacts are most prominent, and was therefore expected to exhibit a noticeable reduction in large amplitude deflections. In contrast, the Oz channel, located in the occipital region, was used as a control to verify that neural signals unrelated to the artifacts remained unaffected.

Following reconstruction, the Fz signal was observed to be substantially cleaner, with previously prominent spike-like artifacts significantly attenuated. Meanwhile, the PSD at the Oz channel remained largely unchanged, confirming that the removal process effectively targeted artifact components without distorting underlying neural activity.

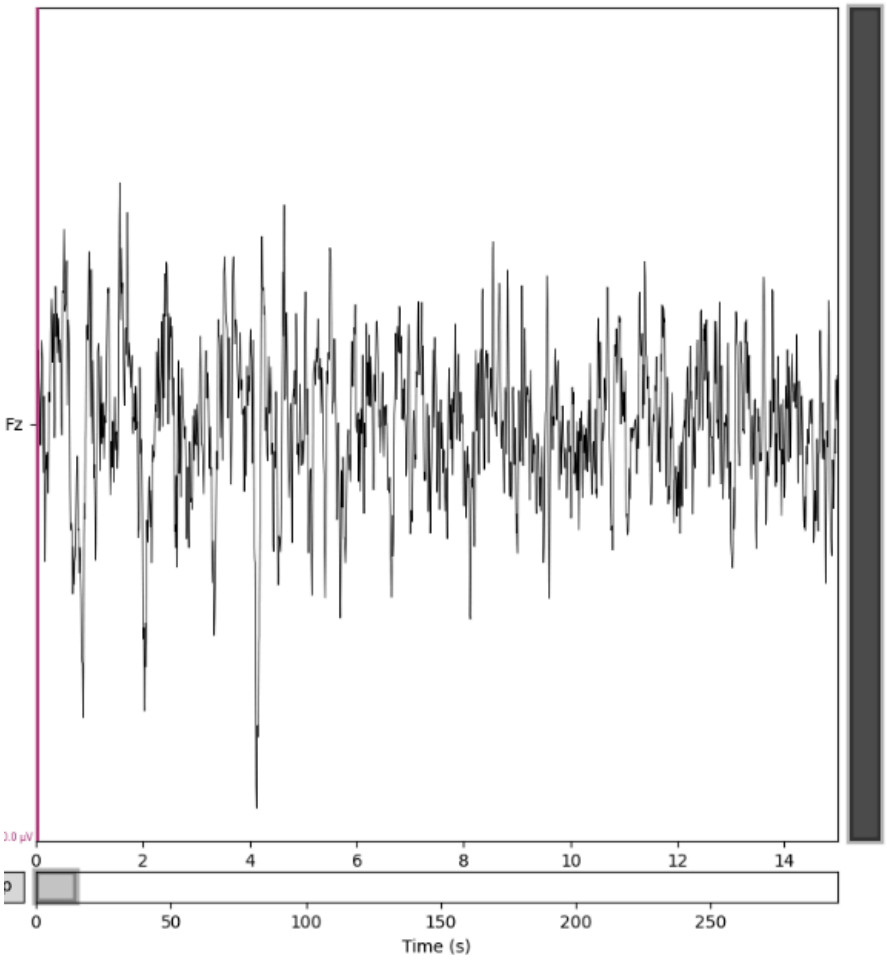


Figure 10. Fz after ICA.

The output confirmed that two components were successfully removed. Following ICA application, the Fz channel appeared substantially smoother, indicating effective attenuation of blink-related artifacts. In contrast, the Oz channel remained largely unchanged, as expected. This outcome demonstrates that ocular artifacts were successfully removed while preserving the underlying neural activity.

Step 10: Overlay plot of Fz

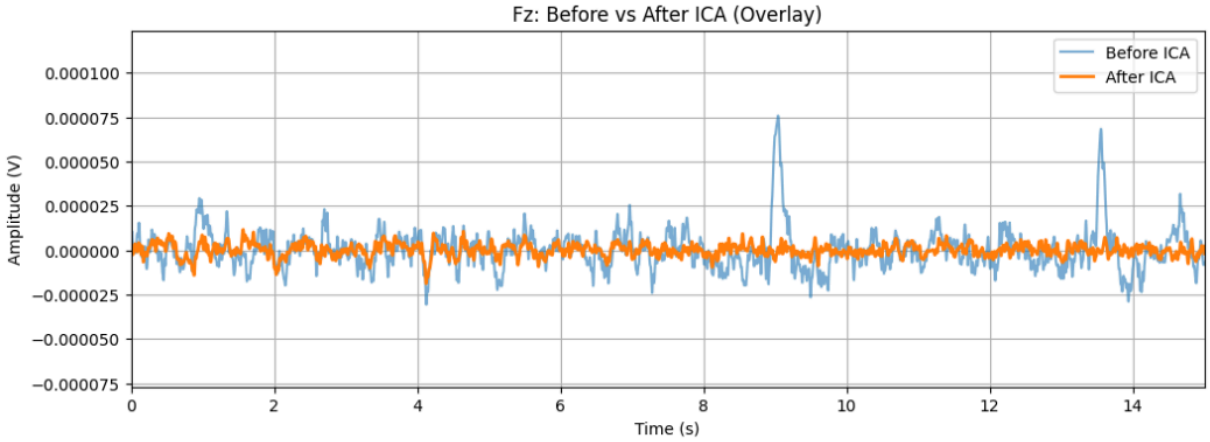


Figure 11. Comparison of Fz before and after ICA.

The Fz signal before and after ICA processing was plotted on the same graph, with the view restricted to the first 15 seconds of data. This comparative visualization facilitates direct assessment of the effect of artifact removal.

Overlaying both signals enables a clear and immediate comparison, highlighting differences attributable to ICA while preserving the overall signal structure. No additional modifications were introduced at this stage, as the plot represents an alternative visualization of the processed and unprocessed signals.

In regions where the pre-ICA signal exhibited large, spike-like deflections, the post-ICA signal appeared substantially smoother, indicating effective removal of blink-related artifacts. In contrast, across other portions of the signal, both traces closely overlapped, demonstrating that ICA selectively removed artifacts without altering the underlying neural activity.

Step 11: Re-epoching and saving

The segmentation of the continuous recording into task-specific blocks and the subsequent division into two-second epochs, as described previously, were repeated following ICA-based

artifact removal. The ±100 microvolt amplitude threshold was then reapplied to reject any remaining contaminated epochs. The resulting cleaned continuous data, along with the updated epochs and corresponding labels, were saved to disk for future use.

Epochs generated prior to ICA processing may still contain residual blink-related artifacts; therefore, reconstruction of the epochs from the cleaned signal was necessary. Persisting the processed data ensures that the preprocessing steps do not need to be repeated in subsequent analyses. As a result of the prior removal of ocular artifacts, a reduction in the number of rejected epochs was expected, reflecting an overall improvement in data quality.

```
Final shape: (250, 8, 500)
Labels:      (array(['eyes_closed_1', 'eyes_closed_2', 'eyes_open_1', 'eyes_open_2',
       'mental_arithmetic'], dtype='<U17'), array([55, 46, 46, 48, 55]))
```

Figure 12. Final Dataset After Full Preprocessing.

The output indicated that 250 epochs were obtained, with a final shape of (250, 8, 500), and block-wise counts of 46, 55, 46, 48, and 55. Notably, these values are identical to those reported in Step 6, which suggests a potential issue in the processing pipeline. Upon inspection of the implementation, it was observed that the segmentation loop references raw_filtered rather than raw_clean. As a result, the reconstructed and saved epochs correspond to the pre-ICA data rather than the artifact-corrected signal. This discrepancy represents a minor but important bug that should be corrected to ensure consistency with the intended preprocessing workflow. Despite this issue, the distribution of epochs across conditions remains relatively balanced, which is advantageous for subsequent analysis.

Step 12: Loading and visualizing the saved data

The previously saved datasets were reloaded for final inspection. Four types of visualizations were generated: a scrolling view of the cleaned EEG signal, the power spectral density (PSD) of the cleaned recording, the average time-domain waveform for each condition, and the average PSD for each condition.

This stage serves as both a final validation step and an initial examination of the experimental outcomes. In particular, the PSD-by-condition visualization provides a basis for evaluating whether distinct cognitive states are associated with measurable differences in neural activity. No further processing was applied at this stage, as the purpose of these visualizations was purely observational. A key feature of interest is the presence of a more pronounced peak around 10 Hz (alpha band activity) during the eyes-closed condition compared to the eyes-open and mental arithmetic conditions. This pattern corresponds to the well-established phenomenon of alpha blocking, in which alpha activity is reduced during visual engagement or cognitive effort.

```
eyes_closed_1       : 55
eyes_closed_2       : 46
eyes_open_1         : 46
eyes_open_2         : 48
mental_arithmetic   : 55
```

Figure 13. Final Dataset.

The output confirmed that the saved datasets were successfully reloaded, with a total of 250 epochs. The distribution across conditions matched the previously stored values, comprising 55 epochs for eyes_closed_1, 46 for eyes_closed_2, 46 for eyes_open_1, 48 for eyes_open_2, and 55 for mental_arithmetic. This consistency indicates that the data loading process was performed correctly.

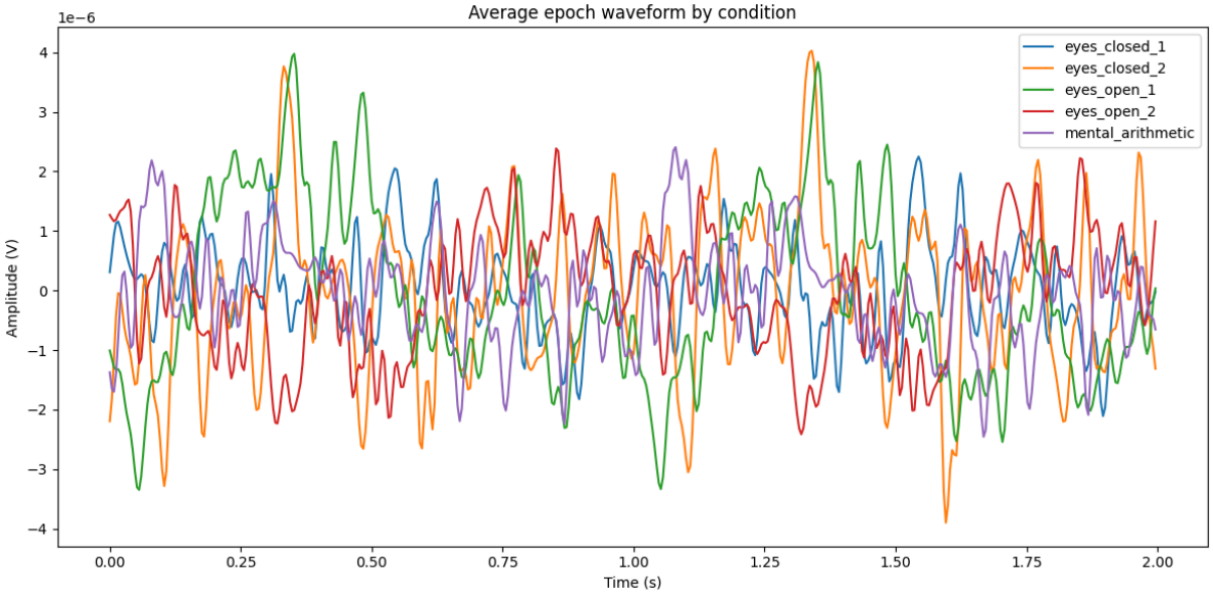


Figure 14. Average epoch waveform.

The average waveform across conditions is expected to appear relatively flat, which is typical for resting-state data, as the signals are not time-locked to specific events. Consequently, this representation provides limited insight compared to frequency-domain analyses.

Greater emphasis is placed on the PSD-by-condition plot. If preprocessing and artifact removal have been effective, the eyes-closed conditions should exhibit a pronounced peak around 10 Hz, corresponding to alpha-band activity. In contrast, the eyes-open and mental arithmetic conditions are expected to show reduced power in this frequency range.

Additionally, close alignment between the two eyes-open conditions would indicate consistency in the participant's behavior across repeated trials. Observation of these patterns supports the

validity of the preprocessing pipeline and suggests that the dataset is of sufficient quality for subsequent analysis stages.

Conclusion

In this study, a five-minute EEG dataset was recorded from a participant using a structured protocol comprising five experimental conditions. The raw data were subsequently processed through a systematic preprocessing pipeline. Initial inspection of the signal revealed the presence of 50 Hz electrical interference and low-frequency drift. To address these issues, band-pass and notch filters were applied, restricting the signal to the frequency range of 1–40 Hz.

Visual inspection of the channels indicated no defective electrodes. The continuous recording was then segmented into two-second epochs, each labeled according to its corresponding condition. Epochs exhibiting excessive peak-to-peak amplitudes were rejected to eliminate prominent artifacts. In addition, Independent Component Analysis (ICA) was employed to identify and remove components associated with eye-blink activity.

The final outcome of this process was a cleaned and segmented dataset consisting of 250 labeled epochs. This dataset is suitable for subsequent analyses, including feature extraction and the classification of different cognitive states.